# Mini/Micro LED Design and Challenges for Advanced Displays

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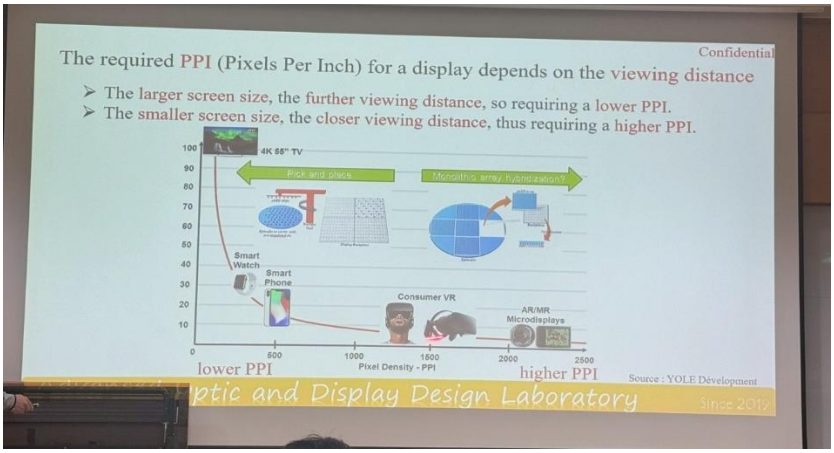
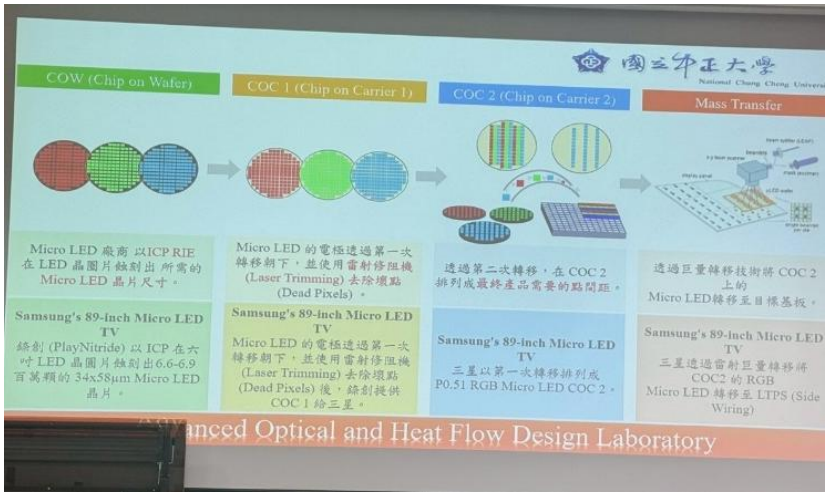
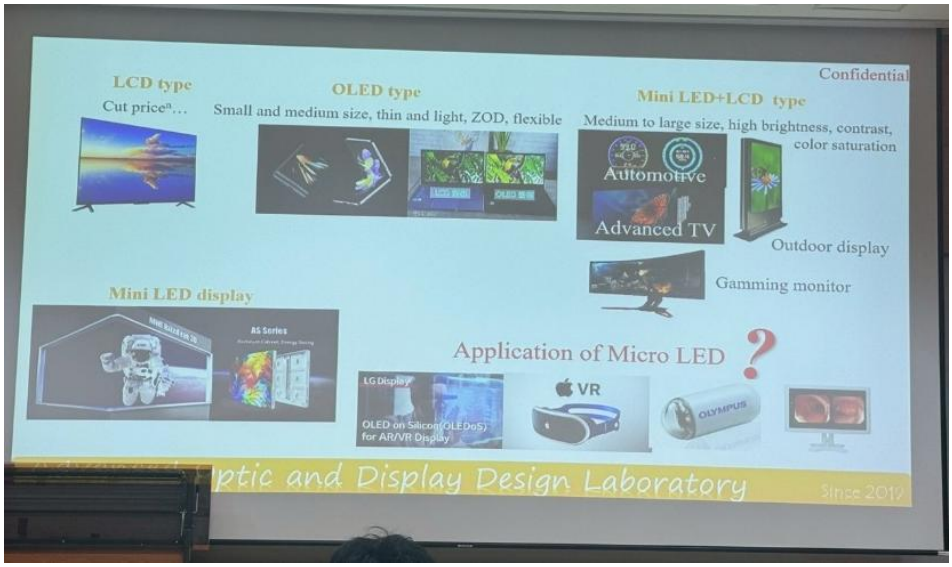



- 顯示技術從 Pick-and-place 轉向 Monolithic array
- 高 PPI 需要更精密製程




1. COW（Chip on Wafer）→ LED 在晶圓上完成
2. COC 1（Chip on Carrier 1）→ 第一次轉移
3. COC 2（Chip on Carrier 2）→ 第二次轉移
4. Mass Transfer → 巨量轉移到最終基板




**1. LCD**

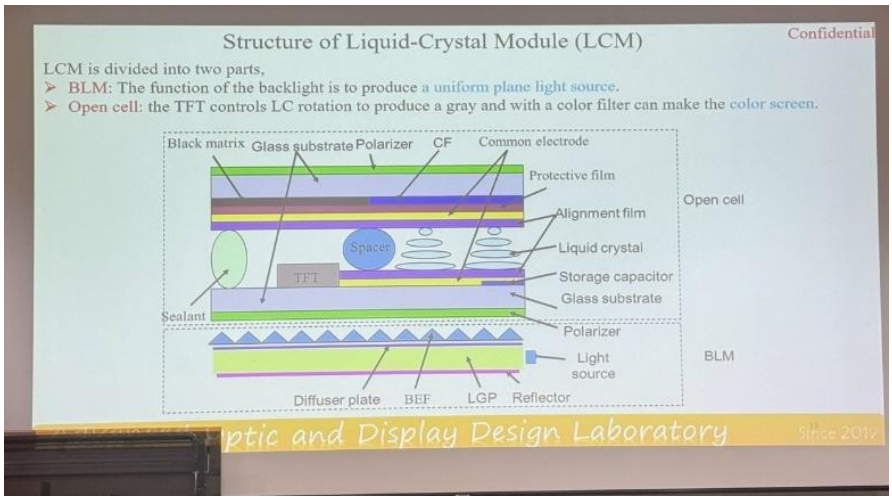
- 價格低
- 傳統背光

**2. OLED**

- 自發光
- 薄、可撓
- 但壽命與亮度受限

**3. Mini LED + LCD**

- 高亮度
- 高對比
- 高色彩飽和

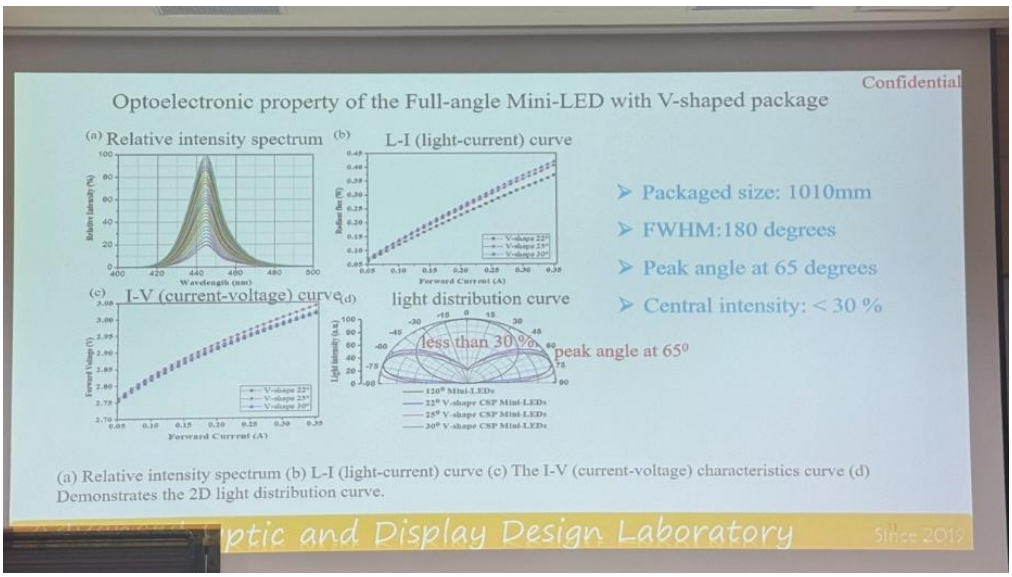



## 1. 背光模組（BLM）

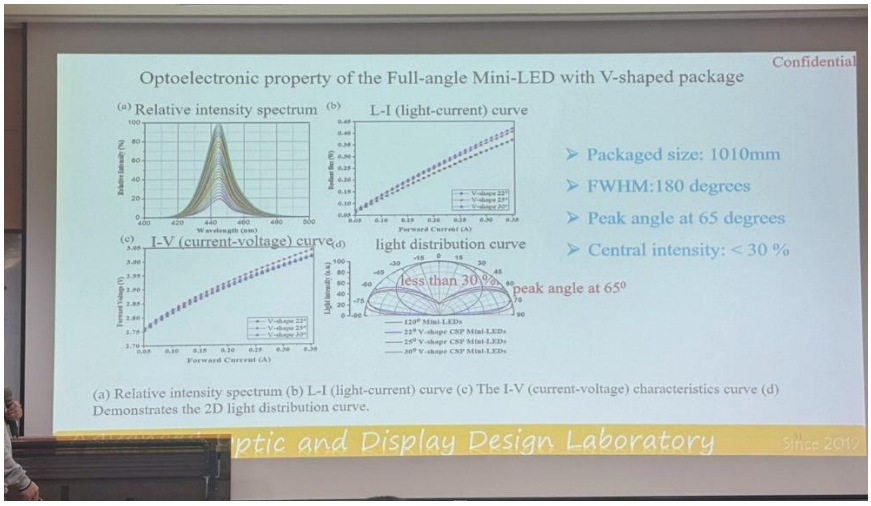
- 產生均勻光源
- Mini LED 就在這裡

## 2. 液晶面板（Open Cell）

- TFT 控制液晶轉動
- 搭配彩色濾光片形成畫面

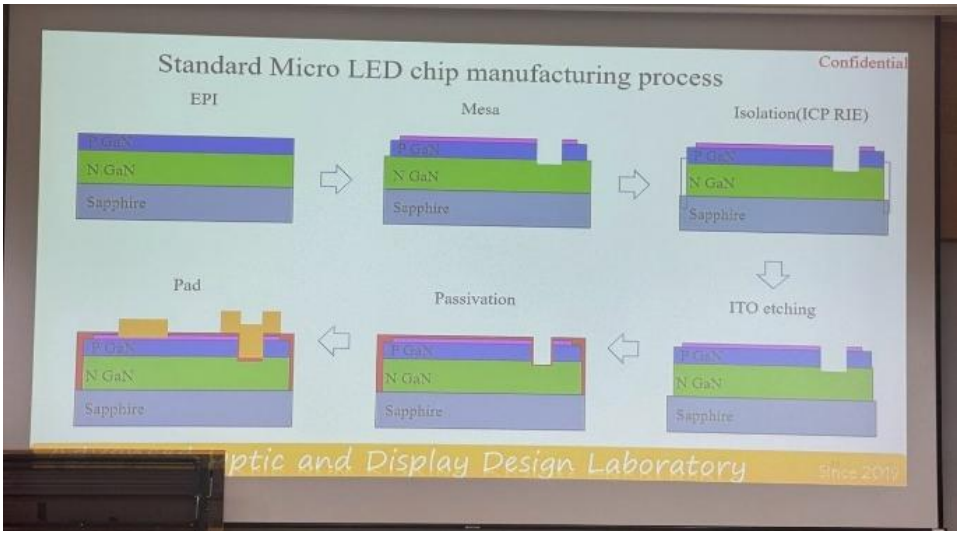



V-shape 讓光更「均勻擴散」，提升背光品質與效率。




V-shape 封裝製程

1. Pick & Place
2. 晶片貼合
3. 白牆層成型（含 $TiO_2$）
4. V 型擴散結構成型
5. 光學層封裝



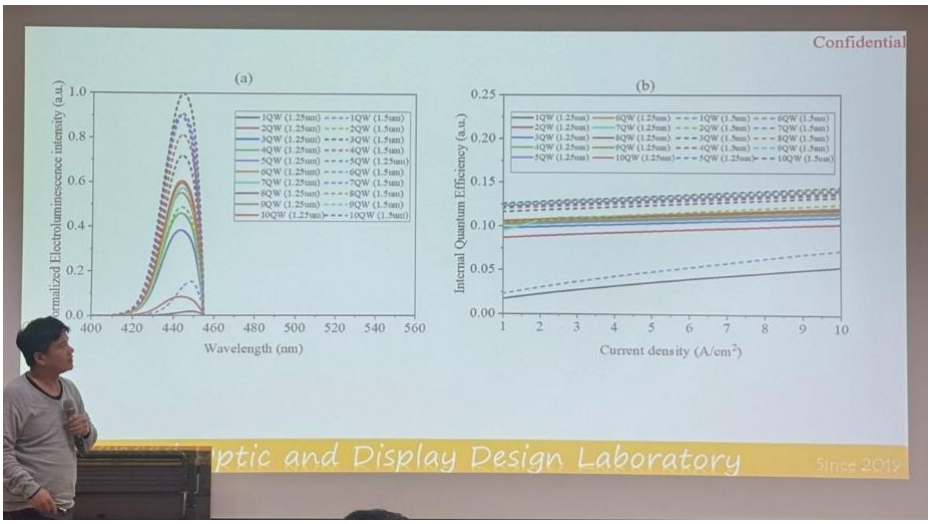

1. EPI 成長（GaN on Sapphire）
2. Mesa 蝕刻

3. ICP 隔離

4. ITO 蝕刻

5. Passivation

6. Pad 製作

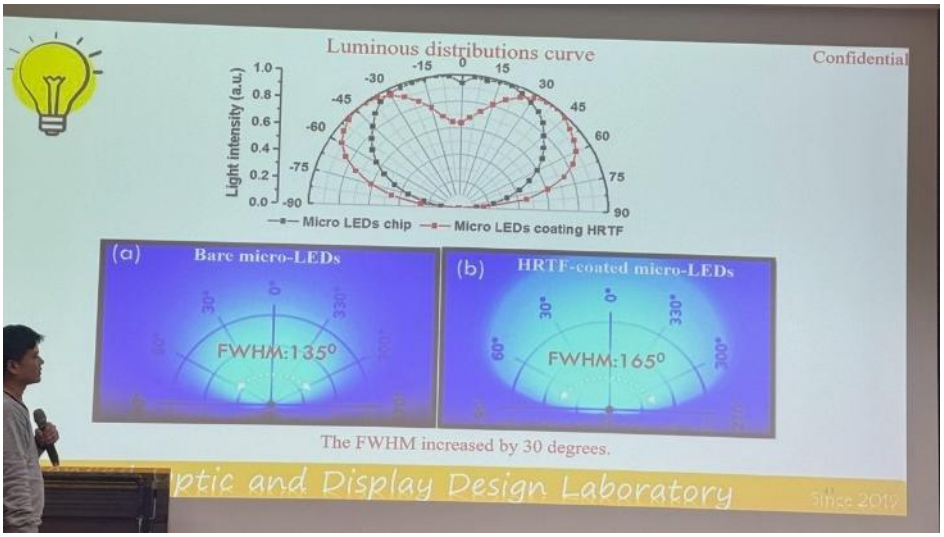



1. **發光光譜（EL spectrum）**

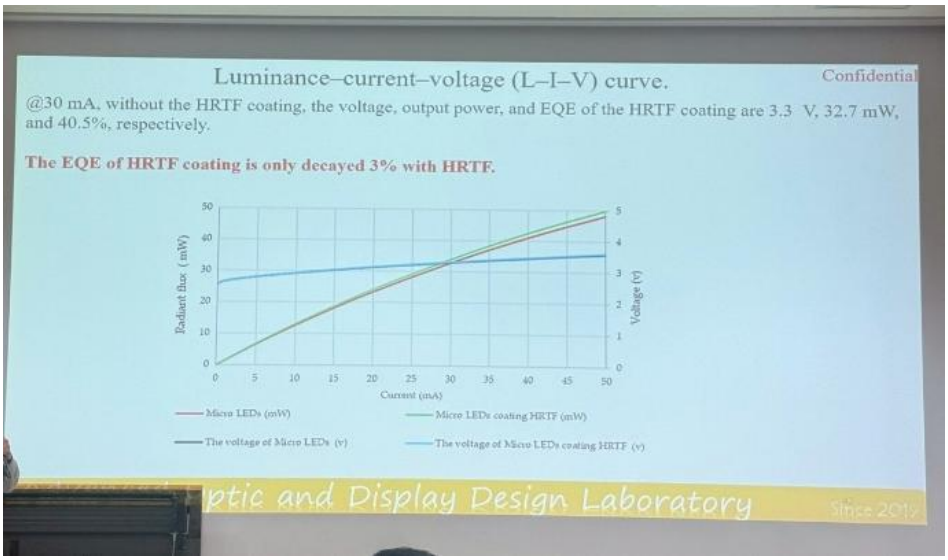
- 不同電流下，主峰約在 440–450 nm
- 強度隨電流增加而提升

2. **亮度效率 vs 電流密度**

- 電流增加 → 亮度提升
- 高電流下效率略下降（效率 droop）



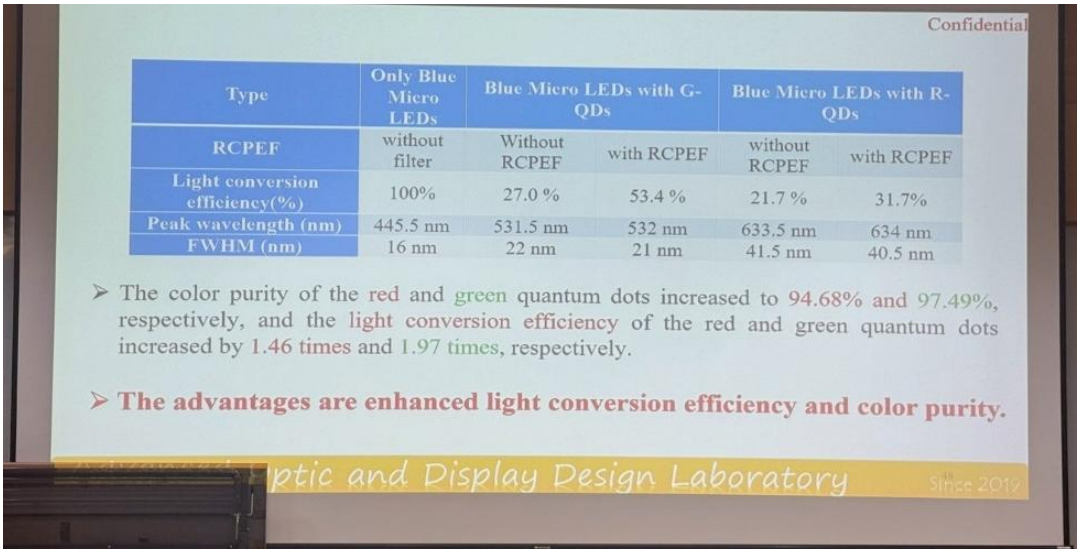

- 裸晶 FWHM：約 135°
- HRTF coating 後：約 165°
- 發光角度增加 30°
- 光分布更均勻




## L–I–V 曲線

- 電流增加 → 亮度線性上升

- 加 HRTF 後：
    - 電壓約 3.3 V
    - 功率約 32.7 mW
- EQE 只下降約 3%



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| Type | Only Blue Micro LEDs | Blue Micro LEDs with G-QDs | | Blue Micro LEDs with R-QDs | |
|---|---|---|---|---|---|
| RCPEF | without filter | Without RCPEF | with RCPEF | without RCPEF | with RCPEF |
| Light conversion efficiency(%) | 100% | 27.0 % | 53.4 % | 21.7 % | 31.7% |
| Peak wavelength (nm) | 445.5 nm | 531.5 nm | 532 nm | 633.5 nm | 634 nm |
| FWHM (nm) | 16 nm | 22 nm | 21 nm | 41.5 nm | 40.5 nm |

➢ The color purity of the red and green quantum dots increased to 94.68% and 97.49%, respectively, and the light conversion efficiency of the red and green quantum dots increased by 1.46 times and 1.97 times, respectively.

➢ **The advantages are enhanced light conversion efficiency and color purity.**

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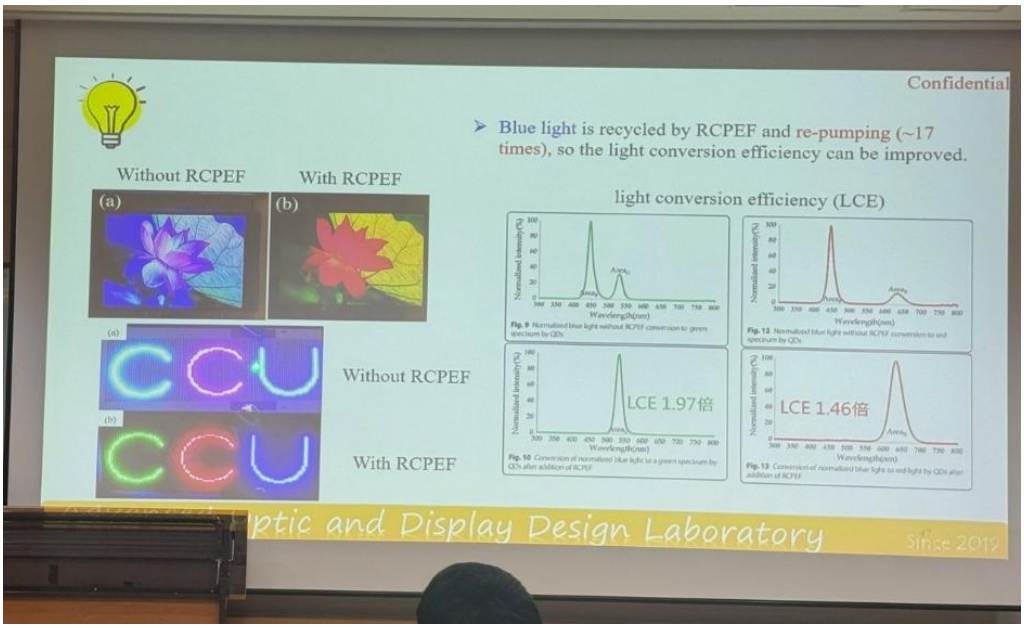
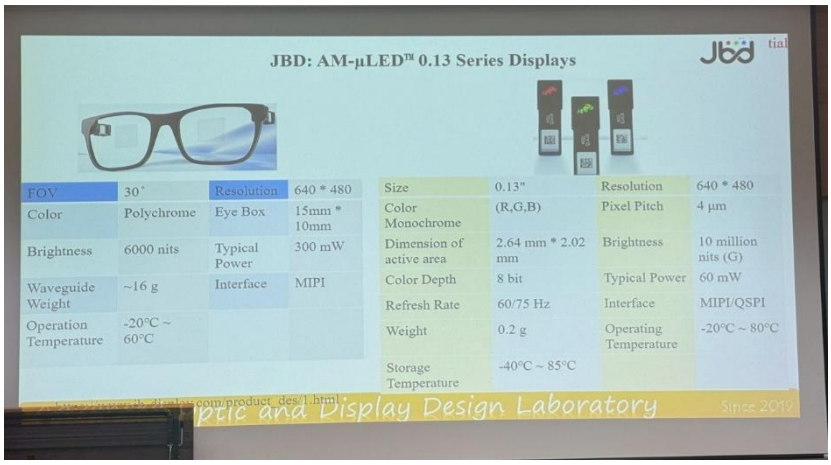
加 R C P E F 後

- 紅、綠 QD 色純度提升到約 95%
- 光轉換效率提升 1.46～1.97 倍



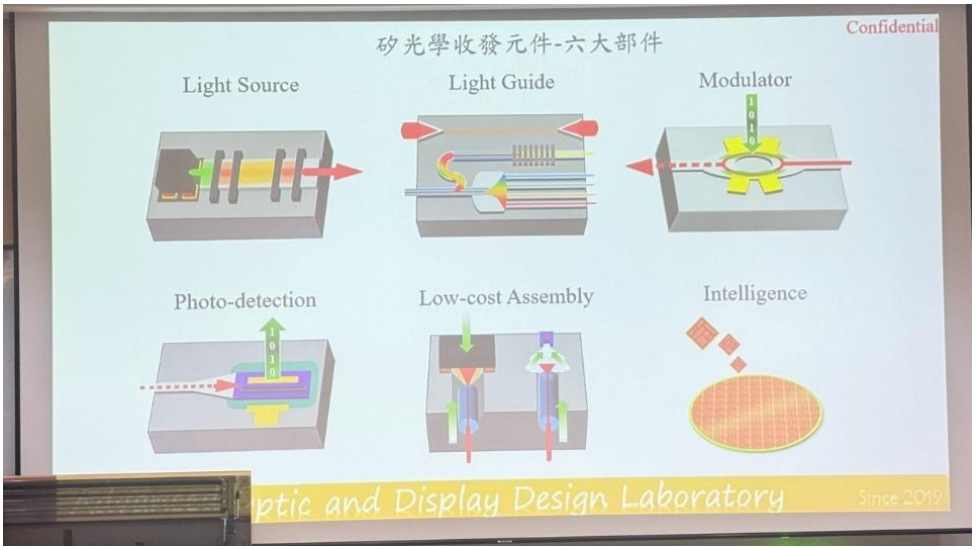

- 藍光被回收並重複激發（re-pumping）
- 光轉換效率明顯提升
- 實驗顯示 LCE 提高 1.46～1.97 倍




## JBD 0.13 吋 AM-μLED 顯示模組

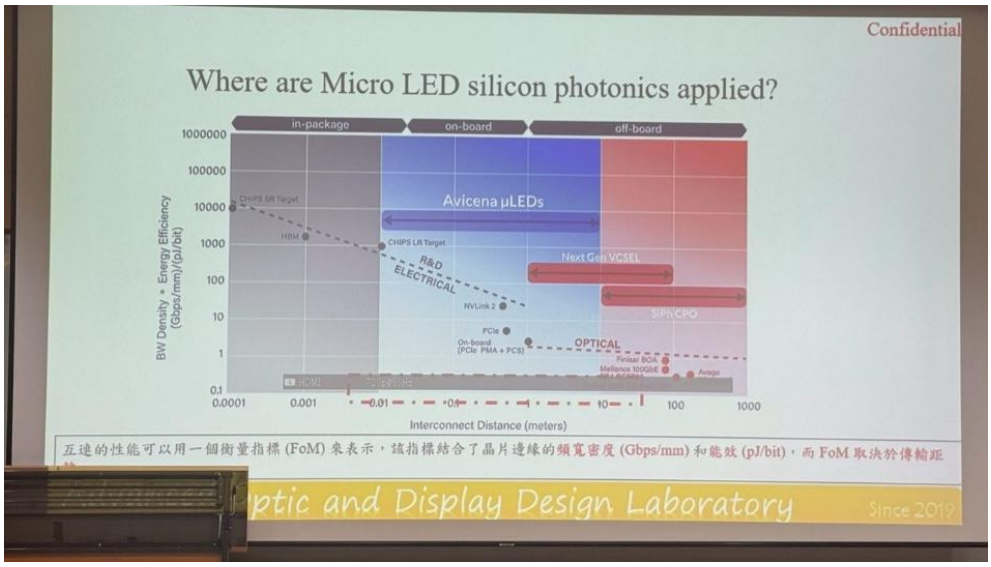
- 解析度：640 × 480
- 像素間距：4 μm

- 亮度：最高 10M nits（G）
- 功耗：約 60–300 mW
- 超小尺寸（0.13 吋）



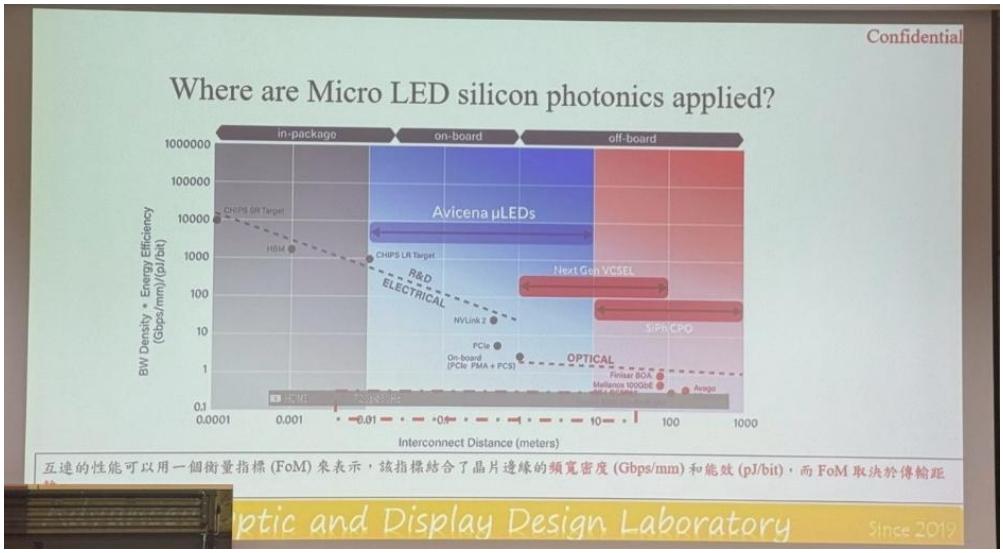

**矽光學六大元件**

1. Light Source（光源）
2. Light Guide（導光）
3. Modulator（調制）
4. Photo-detection（光偵測）
5. Low-cost Assembly（封裝整合）
6. Intelligence（晶片運算）




**Micro LED + Silicon Photonics 應用**

- 用於高速資料傳輸（In-package / On-board / Off-board）
- 光學傳輸優於傳統電傳輸（距離遠、功耗低）
- 應用於資料中心、AI 伺服器



## 市場成長趨勢

- 2022–2027 CAGR 約 48%
- CPO 佔比最高（約 64%）
- 主要應用：資料中心、Optical I/O